Grasshopper-MATLAB Bridge for Lunar Habitat Damage and Repair Simulation

Adelaide University

1. PROBLEM

Manual Transfer

Slow Workflow

Error-prone Setup

The current system requires the user to manually write the Grasshopper 3D model data into the MATLAB HabSim file, run the HamSim virtual simulation, and copy the output data back into Grasshopper. This is laborious, time consuming, and prone to human error and typos.

2. PROJECT AIM

Build a file-based bridge between Grasshopper and HabSim in MATLAB, allowing the transfer and mapping of variables from GH to MATLAB and vice versa, running the HabSim virtual simulation automatically, and notifying the user once the simulation is completed.

3. WORKFLOW

Grasshopper (Model) → Python Bridge → Match & Validate Data → MATLAB / HABSIM → Acquire Results

4. KEY FEATURES

User-friendly GUI

- File Browser
- Custom Configuration
- Change Variables
- Link/Unlink Variables
- Grasshopper Internal Application
- External Application

Fuzzy Matching Algorithm

- Finds Similarly Spelt Variables
- Typo Recognition
- Links Variables with High Confidence
- Helps Identify Spelling Errors

Transfer History

- Stores Previous Transfers
- Old Variables Easily Accessible
- Restore Previous State
- Show Dates and Times of Transfer

Notification System

- Notifies User when Simulation Completed
- Notifications to Desktop
- Notifications to Mobile Device

Supports Strings & Arrays

- Allows for a Variety of Data Types
- Preloaded Configuration with Several Equivalent Data Structures Between Grasshopper and MATLAB
- Allows for Custom Data Equivalency Configurator

Export Results (JSON/TXT/CSV)

- Can Easily Export Data to Any File Type
- Works for Any Input File
- Instant Transfer

5. EXAMPLES

GRASSHOPPER (Various Blocks)

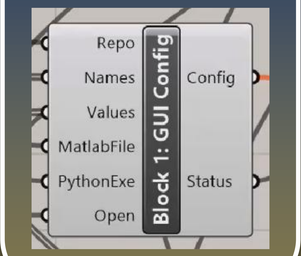
```
Names block:
Length
Width
Height
Material
Supports
Load

Values block:
24.0
8.5
3.0
“steel”
1,0
10,5
```

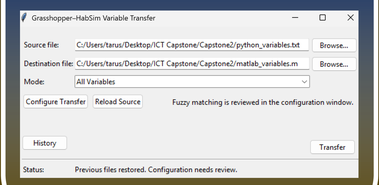
MATLAB / HABSIM (script.m)

```
Length = 24.0;
Width = 8.5;
Height = 3.0;
Material = "steel";
Supports = [1, 0];
Load = [10, 5];
```

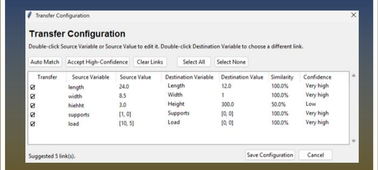
Custom Grasshopper Block



GUI – Main



GUI – Transfer Configuration



6. OUTPUT & IMPACT

Allows for Looping Simulations

Less Manual Editing for Researchers

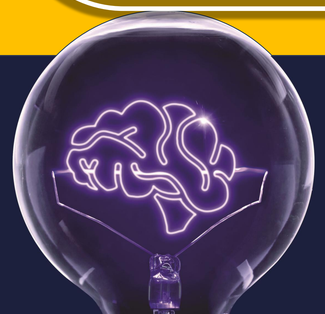
Multi-Way Workflow

7. NEXT STEPS

API Process Automation

Additional I/O Parsing

Extended Analysis & Optimisation



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